MILITARY SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY

TYPES 5FP7A, 5FP14A AND 5FP25

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Magnetic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb1	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0, -180	770	8,800	±125	See note 1	60,000
Minimum:	5.7	---	---	4,000	---	---	---
TEST CONDITIONS:	6.3	Adjust	250	4,000	---	---	---

GENERAL:

Qualification - Required (See note 2)

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
1141	Pressure (implosion)	All		---	---	---	---	---	---
1002	Barometric pressure, reduced	All	Pressure - 54 mmHg	---	---	---	---	---	---
5111	Vibration	All	See note 4	---	---	Width	---	2.0	mm
1331	Direct-interelectrode capacitance	All	k to all g1 to all g2 to all	--- --- ---	--- --- ---	Ck Cg1 Cg2	--- --- ---	10.0 11.0 12.0	pF pF pF
---	Blue/orange light ratio	P14A	See note 5	---	---	Ratio	1.0	1.5	---
5201	Electrode currents (grid No. 2)	All	Ec1 - 0	---	---	Ic2	---	50	µAdc
5101	Face tilt	All		---	---	---	---	---	---
5101	Neck and bulb alinement (magnetic types)	All		---	---	---	---	---	---
5216	Cathode illumination	All		---	---	---	---	---	---
1216	Base material insulating quality	All		---	---	---	---	---	---
	Quality conformance inspection, part 1								
5106	Bulb, screen, and faceplate quality	All		---	---	---	---	---	---
5201	Voltage breakdown	All		---	---	---	---	---	---
5201	Voltage breakdown (magnetic types)	All		---	---	---	---	---	---
5223	Modulation	All	Ib - 200 µAdc	---	---	ΔEc1	---	38	Vdc
5206	Gas ratio	All	See note 3	---	---	Gr	---	0.25	---
5101	Neck straightness	All		---	---	---	---	---	---
---	Neck shadow	P7A P14A	Ib = 50 µAdc (see note 7)	---	---	---	3.82	---	in
5231	Spot position (magnetic deflection)	All		---	---	---	---	9.0	mm
5236	Zero-bias anode current (magnetic deflection)	All		---	---	---	---	---	---
5241	Grid cutoff voltage	All		---	---	Ec1	-25	-70	Vdc
5251	Grid No. 1 leakage current	All		---	---	Ic1	---	3	µAdc
---	Aperture alinement	All	See note 6	---	---	Dis-tance	---	5	mm

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2								
1101	Secureness of base, cap or insert	All		---	---	---	---	---	---
1301	Heater or filament current	All		---	---	If	540	660	mA
5101	Side terminal and base alinement	All		---	---	---	---	---	---
5216	Stray light emission (conventional types)	All P14A P7A P25	Eb = 8,800 Vdc Ec2 = 330 Vdc Ec2 = 770 Vdc Ec2 = 770 Vdc	---	---	---	---	---	---
5221	Screens	All		---	---	---	---	---	---
5226	Line width "A" (magnetic deflection)	P7A P14A P25	Ib = 200 µAdc Ib = 200 µAdc Ib = 25 µAdc	--- --- ---	--- --- ---	Width Width Width	--- --- ---	0.50 0.25 0.20	mm mm mm
5226	Line width "C" (magnetic deflection)	P7A P14A P25	Ib = 200 µAdc Ib = 200 µAdc Ib = 25 µAdc	--- --- ---	--- --- ---	Width Width Width	--- --- ---	0.60 0.35 0.30	mm mm mm
5246	Focusing ampere turns	All	Ib = 200 µAdc; D = 2.75 in.	---	---	---	369	495	---
5251	Heater-cathode leakage current	All		---	---	---	---	---	---
5251	Grid No. 2 leakage current	All		---	---	---	---	---	---
1105	Permanence of marking	All		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life-test provisions	All	Group C; t = 500 hr (min)	---	---	---	---	---	---
		P14A		---	---	---	---	---	---
		P7A	Eb = 8,800 Vdc; Ec2 = 330 Vdc; Ib = 60 µAdc;	---	---	---	---	---	---
		P25	Eb = 8,800 Vdc; Ec2 = 770 Vdc; Ib = 40 µAdc	---	---	---	---	---	---
---	Life-test end points	P7A	Line width "A"	---	---	---	---	0.50	mm
			Line width "C"	---	---	---	---	0.60	mm
		P14A	Line width "A"	---	---	---	---	0.25	mm
			Line width "C"	---	---	---	---	0.35	mm
		P25	Line width "A"	---	---	---	---	0.20	mm
			Line width "C"	---	---	---	---	0.30	mm
		All	Modulation	---	---	ΔEc1	---	38	Vdc
		All	Heater-cathode leakage current	---	---	---	---	---	---
		All	Grid No. 1 leakage current	---	---	---	---	---	---
		All	Grid No. 2 leakage current	---	---	---	---	---	---
		All	Stray light emission	---	---	---	---	---	---

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NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.

2. A limiting aperture type of gun shall be used. The amount of limiting permitted shall be between 15 and 45 percent when Ib = 200 μAdc and the secondary electron current from the screen to the limiting aperture is zero. The following information and materials shall be made available to the government representative witnessing the qualification inspection measurements:

   (a) The electron gun drawing with significant dimensions.

   (b) A sample of the electron gun to be used in the manufacture of the tubes.

   (c) A sample tube with suitable connectors to enable measurement of the screen current separately from the limiting aperture current. This sample tube, at the option of the manufacturer, may be one of the four regular qualification inspection samples or a fifth tube. If the sample is one of the four regular qualification inspection samples, it shall pass all tests. If a fifth tube is submitted, it shall pass all electrical tests, but is not required to pass screen quality, and some of the dimensions.

3. This test to be performed at the conclusion of the holding period.

4. Displacement of the spot corresponding to the image of the final aperture shall not exceed the limit specified.

5. Blue/orange light ratio.

   (a) Using tungsten standard luminant A, calibrate Weston Photronic Cell No. 594-R-R, or equivalent, with Corning 4-65 filter, or equivalent, at 10-foot candles. This combination approximates the normal eye response.

   (b) Set up a 7.1 cm by 7.1 cm defocused raster on the cathode-ray tube under test and adjust the grid potential for a brightness reading of 10-foot lamberts when viewed through a Wratten 15 filter, or equivalent, by the photronic cell and filter previously standardized.

   (c) Without disturbing the photronic cell, replace Wratten 15 and Corning 4-65 filters with Corning 5-59 filter, or equivalent, and again measure the indicated brightness in apparent foot lamberts. The thickness of the filters shall be 2.95 and 5.13 mm, respectively.

   (d) Calculate blue/orange light ratio as follows:

$$\text{Blue/orange light ratio} = \frac{\text{Apparent foot lamberts as measured with Corning 5-59 filter, or equivalent.}}{\text{10-foot lamberts}}$$

6. The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified. Ec1 should not be held at zero for more than approximately 30 seconds to prevent damage to the screen.

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NOTES: -Continued

7. The standard deflection yoke spacer (see figure 2) shall be placed on the tube neck so as to fit tightly between the cone of the glass envelope and the standard 70° deflection yoke, JEDEC-120. Focus coil No. 109 shall be located in the same position with respect to the deflection yoke as for other tests made without the spacer. The electron beam shall be deflected by means of horizontal and vertical deflection waveforms adjusted in magnitude to overscan the useful screen area; if there is no neck shadow the raster shall display 35 to 100 lines. Increase the focus current until the raster passes to a defocused condition with the raster lines overlapping while the ends of the lines become abruptly terminated producing a disk pattern approximately circular in shape having a circumference in sharp focus. Neck shadow will be evident as an outer annular ring or shadow where the electron beam does not reach the screen. The focus coil shall be oriented to make the annular ring symmetrical with the tube face. The minimum diameter of the disk pattern shall be equal to or greater than the specified limit.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MU
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

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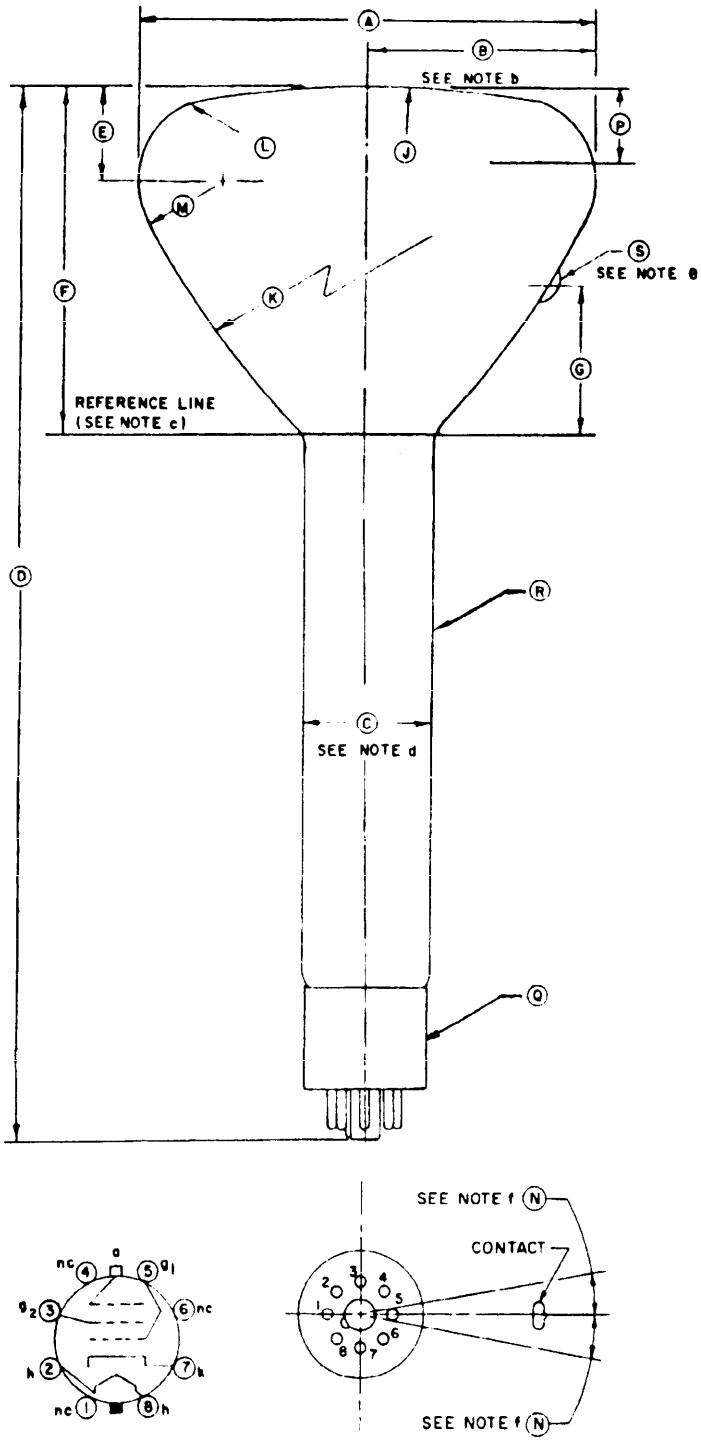


FIGURE 1. Outline drawing of electron tube types 5FP7A, 5FP14A, and 5FP25.

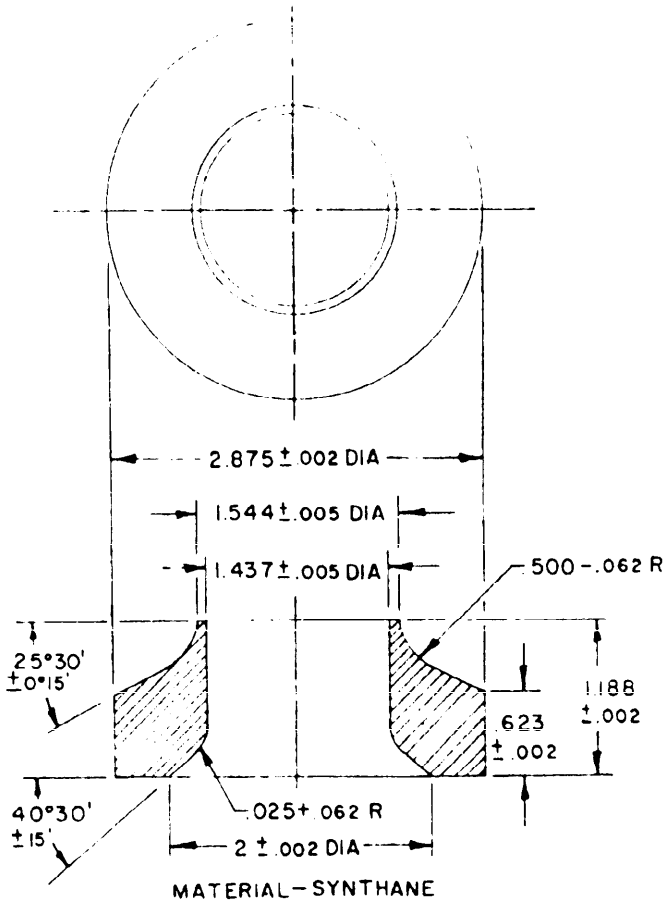
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Type	Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
		Minimum	Maximum
Quality conformance inspection, part 2			
All	A	4.844 (123.04)	5.030 (127.76)
All	B	2.125 (54.98)	
All	C	1.318 (33.48)	1.432 (36.37)
P14A	D	10.75 (273.05)	11.50 (292.10)
P7A, P25	D	10.75 (273.05)	11.125 (282.58)
All	F	3.625 (92.08)	3.749 (95.22)
All	G	1.188 (30.18)	1.562 (39.67)
All	N		10°
Quality conformance inspection, part 3 (periodic check)			
P7A, P25	Q	Base: B8-11, B8-65, B8-26	
P14A	Q	Base: B8-11 or B8-65	
P7A, P25	R	Envelope: J39-½L, J39-½F	
P14A	R	Envelope: J39-½L	
All	S	Contact: J1-22	
Reference dimensions (see note g)			
All	E	1.000 (25.40)	
All	J	24.000 (609.60) R	
All	K	10.000 (254.00) R	
All	L	.125 (3.18) R	
All	M	.906 (23.01) R	
All	P	.812 (20.62)	

NOTES:
(a) All dimensions in inches, unless otherwise specified.
(b) Minimum useful screen radius.
(c) Determined by point where gage G124 will stop.
(d) Applies to entire length of neck below reference line.
(e) Axis of contact terminal shall be normal to bulb surface.
(f) Allowable deviation from true position on center line.
(g) Reference dimensions are for information only and are not required for inspection purposes

FIGURE 1. Outline drawing of electron tube types 5FP7A, 5FP14A, and 5FP25 - Cont.

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NOTE:
All dimensions in inches.

Inches	MM	Inches	MM
.002	.05	1.188	30.18
.005	.13	1.437	36.50
.025	.64	1.544	39.22
.062	1.57	2.000	50.80
.500	12.70	2.875	73.03
.623	15.82		

FIGURE 2. Neck shadow test standard deflection yoke spacer (5FP14A and 5FP7A).

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more, shall be rejected as evidence of air leaks.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 64 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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